

Day 7 — Indian Astronomy

Module: What is IKS? (Middle) · **Time:** 45 min

Learning objective

Students name 3 specific Indian astronomical contributions with their authors/dates and articulate what each got right and what it got wrong.

Materials

- Whiteboard
- Printed extract: 1 verse from *Aryabhaṭīya* in translation
- Globe (if available) for the rotation demo

Vocabulary

- *jyotiṣa* — astronomy/astrology (Vedanga #6)
- *siddhānta* — “established treatise; systematic astronomical text”
- *nakṣatra* — “lunar mansion; star group”

Flow

Warm-up (5 min)

Daily prompt + recall: 2 students name one Vedanga.

Core (25 min)

1. Three specific contributions:

(a) Aryabhata (c. 476–550 CE) — wrote *Aryabhaṭīya* at age 23.

- Proposed the **earth rotates on its axis** (this is correct). Most contemporaries believed the sky rotated.
- Approximated **π to four decimal places** (3.1416).
- Computed planetary periods and eclipse predictions.
- Did NOT propose heliocentrism (the sun-centred model is Copernicus, 1500s).

(b) Sūrya-siddhānta (early CE) — comprehensive astronomical text.

- Detailed planetary calculations.
- Earth’s circumference computed with ~1% accuracy.
- Eclipse predictions accurate enough to publish almanacs.

(c) Varāhamihira (~6th c. CE) — *Pañca-siddhāntikā*. Compares 5 different astronomical systems including a Greek-derived one. Demonstrates that 6th-c. Indian astronomers were actively engaging with Greek (Yavana) astronomical work.

2. What Indian astronomy DID well:

- Precise calendar (lunar + solar reconciliation).
- Eclipse prediction (eclipses were correctly understood as caused by occlusion, not by the demon Rāhu — though the mythological framing co-existed).
- Trigonometric tables (sines, not chords as in Greek tradition).

3. What it DIDN'T do:

- Did not develop heliocentrism (despite Aryabhata's hint, the geocentric model dominated until modern times).
- Did not develop telescopic observation (no telescope until the 17th c. anywhere).

4. The interesting in-between: the Indian astronomical tradition kept astronomy and astrology TOGETHER under jyotiṣa. Calculation skills used to compute eclipses were the same skills used to compute horoscopes. Modern science separates these; the Indian tradition did not.

Activity (10 min) — Aryabhata's earth-rotation argument

Demonstrate with a globe (or a ball with a marked point): - Set the ball spinning. From the spinner's point of view, the room "rotates" around them. - Aryabhata reasoned: maybe the stars don't rotate around us; maybe WE rotate. - This was the correct answer. He was a thousand years ahead of European thinkers on this specific point.

Wrap-up (5 min)

- Preview Day 8: "*Project work begins tomorrow.*"

Student-facing content

Three Indian Astronomical Contributions

1. **Aryabhata (~500 CE)** — proposed earth rotates on its axis; computed π to 4 decimals; predicted eclipses.
2. **Sūrya-siddhānta (early CE)** — planetary calculations + earth's circumference to ~1% accuracy.
3. **Varāhamihira (~6th c. CE)** — *Pañca-siddhāntikā* compared 5 astronomical systems including Greek-derived ones.

Aryabhata's earth-rotation claim was ~1000 years ahead of European thinkers on that specific point.

Note: Indian astronomy did NOT develop heliocentrism (Copernicus, 1500s) or telescopes (17th c.).

Homework

- Finalise project plan. Bring all materials tomorrow.

Differentiation

- **Tier up:** Read about the Yavana (Greek-Indian) interactions in astronomy. Was the influence one-way?
- **Tier down:** Memorise three names with one fact each.

Teacher notes

- The “Aryabhata predicted earth rotation” claim is mainstream and well-supported. Don’t oversell — he didn’t propose heliocentrism.
- The Yavana-influence question is rich: Indian astronomy genuinely engaged Greek astronomy via the *Pañca-siddhāntikā*. Both traditions exchanged ideas. Honest history.

Citation(s) used

- Aryabhaṭīya (499 CE) — Clark (1930) translation.
- Plofker (2009), Chapter 4.
- Pingree, D. (1996). “Astronomy in India” in *Astronomy Before the Telescope*.